

Statistical Errors in Basic Estimates

As noted in Chapter 4, the descriptive properties of a random variable cannot be precisely determined from sample data. Only estimates of the parameters of interest can be obtained from a finite sample of observations. The accuracy of certain basic parameter estimates is discussed in Chapter 4 for the case of data in the form of discrete independent observations of sample size N . In this chapter, the accuracy of parameter estimates is developed for data in the form of continuous time-history records of record length T . It is assumed that the data are single sample records from continuous stationary (ergodic) random processes with arbitrary mean values. Statistical error formulas are developed for

- Mean value estimates
- Mean square value estimates
- Probability density function estimates
- Correlation function estimates
- Autospectral density function estimates

Attention in this chapter and the next chapter is restricted to those errors that are due solely to statistical considerations. Other errors associated with data acquisition and processing are covered in Chapter 10.

8.1 DEFINITION OF ERRORS

Referring to Section 4.1, the accuracy of parameter estimates based on sample values can be described by a mean square error defined as

$$\text{Mean square error} = E[(\hat{\phi} - \phi)^2] \quad (8.1)$$

where $\hat{\phi}$ is an estimator for ϕ . Expanding Equation (8.1) yields

$$\begin{aligned} E[(\hat{\phi} - \phi)^2] &= E[(\hat{\phi} - E[\hat{\phi}] + E[\hat{\phi}] - \phi)^2] \\ &= E[(\hat{\phi} - E[\hat{\phi}])^2] + 2E[(\hat{\phi} - E[\hat{\phi}])(E[\hat{\phi}] - \phi)] \\ &\quad + E[(E[\hat{\phi}] - \phi)^2] \end{aligned}$$

Note that the middle term in the above expression has a factor equal to zero, namely,

$$E[\hat{\phi} - E[\hat{\phi}]] = E[\hat{\phi}] - E[\hat{\phi}] = 0$$

Hence the mean square error reduces to

$$\text{Mean square error} = E[(\hat{\phi} - E[\hat{\phi}])^2] + E[(E[\hat{\phi}] - \phi)^2] \quad (8.2)$$

In words, the mean square error is the sum of two parts. The first part is a variance term that describes the random portion of the error,

$$\text{Var}[\hat{\phi}] = E[(\hat{\phi} - E[\hat{\phi}])^2] = E[\hat{\phi}^2] - E^2[\hat{\phi}] \quad (8.3)$$

and the second part is the square of a bias term that describes the systematic portion of the error,

$$b^2[\hat{\phi}] = E[b^2[\hat{\phi}]] = E[(E[\hat{\phi}] - \phi)^2] \quad (8.4)$$

Thus the mean square error is the sum of the variance of the estimate plus the square of the bias of the estimate, that is,

$$E[(\hat{\phi} - \phi)^2] = \text{Var}[\hat{\phi}] + b^2[\hat{\phi}] \quad (8.5)$$

It is generally more convenient to describe the error of an estimate in terms that have the same engineering units as the parameter being estimated. This can be achieved by taking the positive square roots of the error terms in Equations (8.3)–(8.5). The square root of Equation (8.3) yields the standard deviation for the estimate, called the *standard error* or *random error*, as follows:

$$\text{Random error} = \sigma[\hat{\phi}] = \sqrt{E[\hat{\phi}^2] - E^2[\hat{\phi}]} \quad (8.6)$$

The square root of Equation (8.4) gives the *bias error* directly as

$$\text{Bias error} = b[\hat{\phi}] = E[\hat{\phi}] - \phi \quad (8.7)$$

The square root of the sum of the squared errors, as given by Equation (8.5),

defines the *root-mean-square* (rms) error as

$$\text{rms error} = \sqrt{E[(\hat{\phi} - \phi)^2]} = \sqrt{\sigma^2[\hat{\phi}] + b^2[\hat{\phi}]} \quad (8.8)$$

As a further convenience, it is often desirable to define the error of an estimate in terms of a fractional portion of the quantity being estimated. This is done by dividing the error by the quantity being estimated to obtain a *normalized error*. For $\phi \neq 0$, the normalized random, bias, and rms errors are given by

$$\text{Normalized random error} = \varepsilon_r = \frac{\sigma[\hat{\phi}]}{\phi} = \frac{\sqrt{E[\hat{\phi}^2] - E^2[\hat{\phi}]}}{\phi} \quad (8.9a)$$

$$\text{Normalized bias error} = \varepsilon_b = \frac{b[\hat{\phi}]}{\phi} = \frac{E[\hat{\phi}]}{\phi} - 1 \quad (8.9b)$$

$$\text{Normalized rms error} = \varepsilon = \frac{\sqrt{\sigma^2[\hat{\phi}] + b^2[\hat{\phi}]}}{\phi} = \frac{\sqrt{E[(\hat{\phi} - \phi)^2]}}{\phi} \quad (8.9c)$$

Note that the normalized random error ε_r is often called the *coefficient of variation*.

For situations where ε_r is small, if one sets

$$\hat{\phi}^2 = \phi^2(1 \pm \varepsilon_r)$$

then

$$\hat{\phi} = \phi(1 \pm \varepsilon_r)^{1/2} \approx \phi \left(1 \pm \frac{\varepsilon_r}{2} \right)$$

Thus

$$\varepsilon_r[\hat{\phi}^2] \approx 2\varepsilon_r[\hat{\phi}] \quad (8.10)$$

In words, when ε_r is small, the normalized random error for squared estimates $\hat{\phi}^2$ is approximately twice the normalized random error for unsquared estimates $\hat{\phi}$.

When estimates $\hat{\phi}$ have a negligible bias error $b[\hat{\phi}] \approx 0$ and a small normalized rms error $\varepsilon = \sigma[\hat{\phi}]/\phi$, say $\varepsilon \leq 0.20$, then the probability density function $p(\hat{\phi})$ for these estimates can be approximated by a Gaussian distribution, where the mean value $E[\hat{\phi}] \approx \phi$ and the standard deviation $\sigma[\hat{\phi}] = \varepsilon\phi$, as follows:

$$p(\hat{\phi}) \approx \frac{1}{\varepsilon\phi\sqrt{2\pi}} \exp \left[-\frac{(\hat{\phi} - \phi)^2}{2(\varepsilon\phi)^2} \right] \quad (8.11)$$

This gives probability statements for future values of the estimates $\hat{\phi}$, such as

$$\begin{aligned} \text{Prob}[\phi(1 - \varepsilon) \leq \hat{\phi} \leq \phi(1 + \varepsilon)] &\approx 0.68 \\ \text{Prob}[\phi(1 - 2\varepsilon) \leq \hat{\phi} \leq \phi(1 + 2\varepsilon)] &\approx 0.95 \end{aligned} \quad (8.12)$$

A confidence interval for the unknown true value ϕ based on any single estimate $\hat{\phi}$ then becomes

$$\begin{aligned} \left[\frac{\hat{\phi}}{1 + \varepsilon} \leq \phi \leq \frac{\hat{\phi}}{1 - \varepsilon} \right] &\quad \text{with 68\% confidence} \\ \left[\frac{\hat{\phi}}{1 + 2\varepsilon} \leq \phi \leq \frac{\hat{\phi}}{1 - 2\varepsilon} \right] &\quad \text{with 95\% confidence} \end{aligned} \quad (8.13)$$

For small ε , say $\varepsilon \leq 0.10$, this simplifies to

$$\begin{aligned} [\hat{\phi}(1 - \varepsilon) \leq \phi \leq \hat{\phi}(1 + \varepsilon)] &\quad \text{with 68\% confidence} \\ [\hat{\phi}(1 - 2\varepsilon) \leq \phi \leq \hat{\phi}(1 + 2\varepsilon)] &\quad \text{with 95\% confidence} \end{aligned} \quad (8.14)$$

These confidence claims can be made when ε is small even though the actual unknown sampling distribution for $\hat{\phi}$ is theoretically chi-square, F , or some other more complicated distribution detailed in Chapter 4.

For cases where normalized rms errors are not small, confidence interval statements can still be made, as described in Section 4.4. Such matters will be discussed for spectral density estimates later in this chapter. Along with the derivation of errors, the consistency of various estimates will also be noted, as defined by Equation (4.7).

Example 8.1. Approximate 95% Confidence Intervals for Mean Square and rms Value Estimates. Assume the mean square value of a signal $x(t)$ is estimated to be $\hat{\psi}_x^2 = 4$ with a normalized random error of $\varepsilon_r = 0.05$. Determine the approximate 95% confidence intervals for the mean square value ψ_x^2 and the rms value ψ_x of the signal.

From Equation (8.14), the approximate 95% confidence interval for ψ_x^2 is

$$[4(0.90) \leq \psi_x^2 \leq 4(1.10)] = [3.6 \leq \psi_x^2 \leq 4.4]$$

Using the result in Equation (8.10), $\varepsilon_r = 0.025$ for $\hat{\psi}_x = 2$, so the approximate 95% confidence interval for ψ_x is

$$[2(0.95) \leq \psi_x \leq 2(1.05)] = [1.9 \leq \psi_x \leq 2.1]$$

Note that the confidence interval for ψ_x is approximately the square root of the interval for ψ_x^2 .

8.2 MEAN AND MEAN SQUARE VALUE ESTIMATES

8.2.1 Mean Value Estimates

Suppose that a single sample time-history record $x(t)$ from a stationary (ergodic) random process $\{x(t)\}$ exists over a finite time T . The mean value of $\{x(t)\}$ can be estimated by

$$\hat{\mu}_x = \frac{1}{T} \int_0^T x(t) dt \quad (8.15)$$

The true mean value is

$$\mu_x = E[x(t)] \quad (8.16)$$

and is independent of t because $\{x(t)\}$ is stationary. The expected value of the estimate $\hat{\mu}_x$ is

$$E[\hat{\mu}_x] = E\left[\frac{1}{T} \int_0^T x(t) dt\right] = \frac{1}{T} \int_0^T E[x(t)] dt = \frac{1}{T} \int_0^T \mu_x dt = \mu_x \quad (8.17)$$

because expected values commute with linear operations. Hence $\hat{\mu}_x$ is an unbiased estimate of μ_x , independent of T . Because $\hat{\mu}_x$ is unbiased, the mean square error of the estimate $\hat{\mu}_x$ is equal to the variance as follows:

$$\text{Var}[\hat{\mu}_x] = E[(\hat{\mu}_x - \mu_x)^2] = E[\hat{\mu}_x^2] - \mu_x^2 \quad (8.18)$$

where from Equation (8.15),

$$E[\hat{\mu}_x^2] = \frac{1}{T^2} \int_0^T \int_0^T E[x(\xi)x(\eta)] d\eta d\xi \quad (8.19)$$

Now the autocorrelation function $R_{xx}(\tau)$ of a stationary random process $\{x(t)\}$ is defined by Equation (5.6) as

$$R_{xx}(\tau) = E[x(t)x(t + \tau)] \quad (8.20)$$

From the stationary hypothesis, $R_{xx}(\tau)$ is independent of t , and an even function of τ with a maximum at $\tau = 0$. It will be assumed that $R_{xx}(\tau)$ is continuous and finite for all values of τ , and that all periodic components in $R_{xx}(\tau)$ have been removed at the onset. The autocovariance function $C_{xx}(\tau)$ is defined by Equation (5.8) as

$$C_{xx}(\tau) = R_{xx}(\tau) - \mu_x^2 \quad (8.21)$$

It turns out that whenever $\mu_x \neq 0$, it is more convenient to work with $C_{xx}(\tau)$ than with $R_{xx}(\tau)$. It will be assumed that $C_{xx}(\tau)$ satisfies the integrable properties of Equation (5.123) so as to make $\{x(t)\}$ ergodic.

In terms of the autocovariance function $C_{xx}(\tau)$, the variance (mean square error) from Equations (8.18) and (8.19) becomes

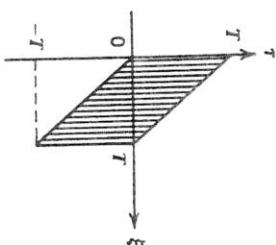
$$\begin{aligned} \text{Var}[\hat{\mu}_x] &= \frac{1}{T^2} \int_0^T \int_0^T C_{xx}(\eta - \xi) d\eta d\xi = \frac{1}{T^2} \int_0^T \int_{-\xi}^{T-\xi} C_{xx}(\tau) d\tau d\xi \\ &= \frac{1}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) C_{xx}(\tau) d\tau \end{aligned} \quad (8.22)$$

The last expression occurs by reversing the orders of integration between τ and ξ and carrying out the ξ integration. This changes the limits of integration for τ and ξ as shown in the sketch below so that

$$\begin{aligned} \int_0^T \int_{-\xi}^{T-\xi} C_{xx}(\tau) d\tau d\xi &= \int_{-T}^0 \int_{-T}^T C_{xx}(\tau) d\xi d\tau + \int_0^T \int_0^{T-\tau} C_{xx}(\tau) d\xi d\tau \\ &= \int_{-T}^0 (T + \tau) C_{xx}(\tau) d\tau + \int_0^T (T - \tau) C_{xx}(\tau) d\tau \\ &= \int_{-T}^T (T - |\tau|) C_{xx}(\tau) d\tau \end{aligned}$$

Now on letting T tend to infinity, Equation (8.22) becomes

$$\lim_{T \rightarrow \infty} T \text{Var}[\hat{\mu}_x] = \int_{-\infty}^{\infty} C_{xx}(\tau) d\tau < \infty \quad (8.23)$$



where Equation (5.122) is used, which provides that $C_{xx}(\tau)$ and $\tau C_{xx}(\tau)$ are absolutely integrable over $(-\infty, \infty)$ to justify passage to the limit inside the integral sign. In particular, Equation (8.23) shows that for large T , where $|\tau| \ll T$, the variance is given by

$$\text{Var}[\hat{\mu}_x] \approx \frac{1}{T} \int_{-\infty}^{\infty} C_{xx}(\tau) d\tau \quad (8.24)$$

Hence when the integral is finite-valued, $\text{Var}[\hat{\mu}_x]$ approaches zero as T approaches infinity proving that $\hat{\mu}$ is a consistent estimate of μ .

Consider the important special case where $\{x(t)\}$ is *bandwidth-limited white noise* with a mean value $\mu_x \neq 0$ and a variance σ_x^2 . Its autospectral density function can be described by

$$G_{xx}(f) = \begin{cases} \frac{\sigma_x^2}{B} + \mu_x^2 \delta(f) & 0 \leq f \leq B \\ 0 & f > B \end{cases} \quad (8.25)$$

where B is the bandwidth. The associated autocovariance function is given by

$$\begin{aligned} C_{xx}(\tau) &= \int_0^\infty G_{xx}(f) \cos 2\pi f\tau \, df - \mu_x^2 \\ &= \sigma_x^2 \left(\frac{\sin 2\pi B\tau}{2\pi B\tau} \right) \end{aligned} \quad (8.26)$$

Note that $C_{xx}(0) = \sigma_x^2$ and $C_{xx}(\tau) = 0$ for $\tau = (n/2B)$, where n is an integer. Thus points $(1/2B)$ apart are uncorrelated. They will be statistically independent if $\{x(t)\}$ is also Gaussian. For this case, Equation (8.24) yields the approximate result when $BT \geq 5$ that

$$\text{Var}[\hat{\mu}_x] \approx \frac{\sigma_x^2}{2BT} \quad (8.27)$$

When $\mu_x \neq 0$, the normalized rms error is

$$e[\hat{\mu}_x] \approx \frac{1}{\sqrt{2BT}} \left(\frac{\sigma_x}{\mu_x} \right) \quad (8.28)$$

From Equation (8.3), note that

$$E[\hat{\mu}_x^2] = \text{Var}[\hat{\mu}_x] + \mu_x^2 \quad (8.29)$$

Hence, using Equation (8.27),

$$E[\hat{\mu}_x^2] \approx \frac{\sigma_x^2}{2BT} + \mu_x^2 \quad (8.30)$$

For Gaussian data, when $\mu_x \neq 0$, the fourth-order moment from Equation (3.82) becomes

$$E[\hat{\mu}_x^4] = 3\{E[\hat{\mu}_x^2]\}^2 - 2\mu_x^4 \quad (8.31)$$

Then, neglecting terms of order $(1/BT)^2$,

$$\text{Var}[\hat{\mu}_x^2] = E[\hat{\mu}_x^4] - \{E[\hat{\mu}_x^2]\}^2 \approx \frac{2\mu_x^2\sigma_x^2}{BT} \quad (8.32)$$

Thus the normalized mean square random error for bandwidth-limited Gaussian white noise is given by

$$e_r^2[\hat{\mu}_x^2] = \frac{\text{Var}[\hat{\mu}_x^2]}{\mu_x^4} \approx \frac{2}{BT} \left(\frac{\sigma_x}{\mu_x} \right)^2$$

Now, comparing with Equation (8.28),

$$e_r^2[\hat{\mu}_x^2] \approx 4e^2[\hat{\mu}_x]$$

and

$$e_r[\hat{\mu}_x^2] \approx 2e[\hat{\mu}_x] \approx \frac{\sqrt{2}}{\sqrt{BT}} \left(\frac{\sigma_x}{\mu_x} \right) \quad (8.33)$$

This agrees with the general relation of Equation (8.10).

Example 8.2. Random Error in Mean Value Estimate. Consider a bandwidth-limited white noise signal $x(t)$ with a bandwidth of $B = 100$ Hz, a mean value of $\mu_x = 0$, and a standard deviation of $\sigma_x = 2$. If the mean value of $x(t)$ is to be estimated by averaging over a record of length $T = 2$ sec, determine an interval that will include the mean value estimate with a probability of approximately 95%.

From Equation (8.27), the random error of the estimate $\hat{\mu}_x$ will have a standard deviation of

$$\sigma[\hat{\mu}_x] = \frac{2}{\sqrt{2(100)(2)}} = 0.10$$

It then follows from Equation (8.12) that the probability is about 95% that an estimate $\hat{\mu}_x$ will fall within the interval

$$[-0.2 \leq \hat{\mu}_x \leq 0.2]$$

8.2.2 Mean Square Value Estimates

As in Section 8.2.1, let $x(t)$ be a single-sample time-history record from a stationary (ergodic) random process $\{x(t)\}$. The mean square value of $\{x(t)\}$ can be estimated by time averaging over a finite time interval T as follows:

$$\hat{\psi}_x^2 = \frac{1}{T} \int_0^T x^2(t) \, dt \quad (8.34)$$

The true mean square value is

$$\psi_x^2 = E[x^2(t)] \quad (8.35)$$

and is independent of t because $\{x(t)\}$ is stationary. The expected value of the estimate $\hat{\psi}_x^2$ is

$$E[\hat{\psi}_x^2] = \frac{1}{T} \int_0^T E[x^2(t)] dt = \frac{1}{T} \int_0^T \psi_x^2 dt = \psi_x^2 \quad (8.36)$$

Hence $\hat{\psi}_x^2$ is an unbiased estimate of ψ_x^2 , independent of T .

The mean square error here is given by the variance

$$\begin{aligned} \text{Var}[\hat{\psi}_x^2] &= E[(\hat{\psi}_x^2 - \psi_x^2)^2] = E[\hat{\psi}_x^4] - \psi_x^4 \\ &= \frac{1}{T^2} \int_0^T \int_0^T (E[x^2(\xi)x^2(\eta)] - \psi_x^4) d\eta d\xi \end{aligned} \quad (8.37)$$

Assume now that $\{x(t)\}$ is a Gaussian random process with a mean value $\mu_x \neq 0$. Then the expected value in Equation (8.37) takes the special form, derived from Equation (3.81), as follows:

$$E[x^2(\xi)x^2(\eta)] = 2(R_{xx}^2(\eta - \xi) - \mu_x^4) + \psi_x^4 \quad (8.38)$$

From the basic relationship in Equation (8.21), we obtain

$$R_{xx}^2(\eta - \xi) - \mu_x^4 = C_{xx}^2(\eta - \xi) + 2\mu_x^2 C_{xx}(\eta - \xi) \quad (8.39)$$

Hence

$$\begin{aligned} \text{Var}[\hat{\psi}_x^2] &= \frac{2}{T^2} \int_0^T \int_0^T (R_{xx}^2(\eta - \xi) - \mu_x^4) d\eta d\xi \\ &= \frac{2}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) (R_{xx}^2(\tau) - \mu_x^4) d\tau \\ &= \frac{2}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) (C_{xx}^2(\tau) + 2\mu_x^2 C_{xx}(\tau)) d\tau \end{aligned} \quad (8.40)$$

For large T , where $|\tau| \ll T$, the variance becomes

$$\text{Var}[\hat{\psi}_x^2] \approx \frac{2}{T} \int_{-\infty}^{\infty} (C_{xx}^2(\tau) + 2\mu_x^2 C_{xx}(\tau)) d\tau \quad (8.41)$$

Thus $\hat{\psi}_x^2$ is a consistent estimate of ψ_x^2 because $\text{Var}[\hat{\psi}_x^2]$ will approach zero as T approaches infinity, assuming that, $C_{xx}^2(\tau)$ and $C_{xx}(\tau)$ are absolutely integrable over $(-\infty, \infty)$, as stated in Equation (5.122).

Consider the special case of bandwidth-limited Gaussian white noise as defined by Equation (8.25). From Equation (8.26).

$$C_{xx}(\tau) = \sigma_x^2 \left(\frac{\sin 2\pi B\tau}{2\pi B\tau} \right) \quad (8.42)$$

For this case, Equation (8.41) shows that

$$\text{Var}[\hat{\psi}_x^2] \approx \frac{\sigma_x^4}{BT} + \frac{2}{BT} \mu_x^2 \sigma_x^2 \quad (8.43)$$

This is the variance of mean square value estimates, where B is the total bandwidth of the data and T is the total record length of the data. In general, for $\mu_x \neq 0$, the normalized rms error is

$$e[\hat{\psi}_x^2] = \frac{\text{s.d.}[\hat{\psi}_x^2]}{\psi_x^2} \approx \frac{1}{\sqrt{BT}} \left(\frac{\sigma_x}{\psi_x} \right)^2 + \frac{\sqrt{2}}{\sqrt{BT}} \left(\frac{\mu_x \sigma_x}{\psi_x^2} \right) \quad (8.44)$$

For those cases where $\mu_x = 0$, the quantity $\psi_x^2 = \sigma_x^2$ and results simplify to

$$\text{Var}[\hat{\psi}_x^2] \approx \frac{\psi_x^4}{BT} = \frac{R_{xx}^2(0)}{BT} \quad (8.45)$$

with

$$e[\hat{\psi}_x^2] \approx \frac{1}{\sqrt{BT}} \quad (8.46)$$

Corresponding results for rms value estimates $\hat{\psi}_x$ instead of $\hat{\psi}_x^2$, when $\mu_x = 0$, are found from Equation (8.10) to be

$$e[\hat{\psi}_x] \approx \frac{1}{2\sqrt{BT}} \quad (8.47)$$

Plots of Equations (8.46) and (8.47) versus the BT product are presented in Figure 8.1.

Again for cases where $\mu_x = 0$, the normalized random error for mean square value estimates, as given for bandwidth-limited white noise in Equation (8.46), can be generalized for Gaussian random data with any arbitrary autospectral density function, as follows. From Parseval's theorem in Equation (5.75),

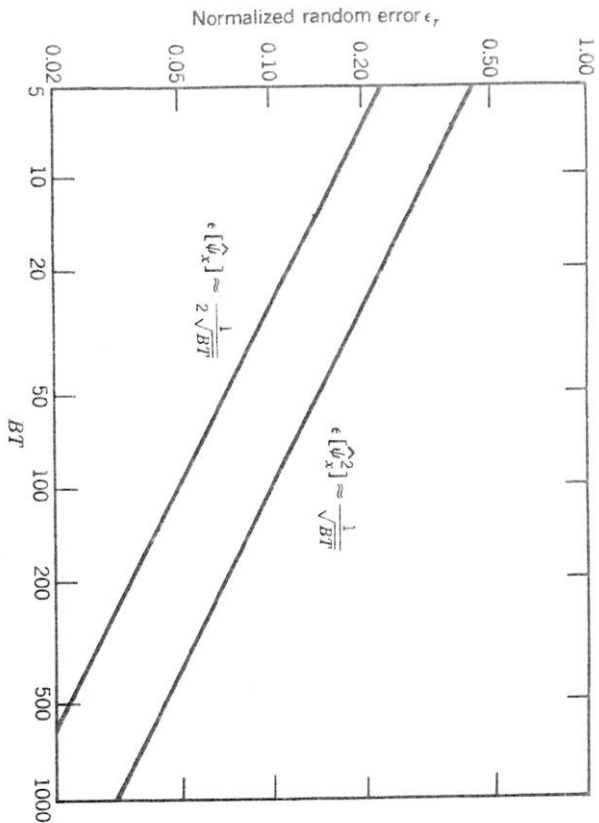


Figure 8.1 Normalized random error of mean square value and rms value estimates.

because $S_{xx}(f)$ is the Fourier transform of $C_{xx}(f)$ and $G_{xx}(f) = 2S_{xx}(f)$ when $f > 0$, it follows that

$$\int_{-\infty}^{\infty} C_{xx}^2(\tau) d\tau = \int_{-\infty}^{\infty} S_{xx}^2(f) df = 2 \int_0^{\infty} S_{xx}^2(f) df = \frac{1}{2} \int_0^{\infty} G_{xx}^2(f) df \quad (8.48)$$

Substituting Equation (8.48) into Equation (8.41) with $\mu_x = 0$ yields

$$\text{Var}[\hat{\psi}_x^2] = \frac{1}{T} \int_0^{\infty} G_{xx}^2(f) df \quad (8.49)$$

Hence, the normalized variance of the mean square value estimate becomes

$$\varepsilon_r^2[\hat{\psi}_x^2] = \frac{\text{Var}[\hat{\psi}_x^2]}{\hat{\psi}_x^4} = \frac{\frac{1}{T} \int_0^{\infty} G_{xx}^2(f) df}{\hat{\psi}_x^4} \quad (8.50)$$

and the normalized random error may be written as

$$\varepsilon_r[\hat{\psi}_x^2] = \frac{1}{\sqrt{B_s T}} \quad (8.51)$$

where

$$B_s = \frac{\psi_x^4}{\int_0^{\infty} G_{xx}^2(f) df} = \frac{[\int_0^{\infty} G_{xx}(f) df]^2}{\int_0^{\infty} G_{xx}^2(f) df} \quad (8.52)$$

is called the *statistical bandwidth* of the data. The statistical bandwidth B_s should not be confused with the noise spectral bandwidth B_n defined in Equation (5.106), which is the bandwidth of a stationary random process with an arbitrary autospectral density function that has the same mean square value as white noise with a bandwidth $B = B_n$. Instead, the statistical bandwidth is the bandwidth of a stationary random process with an arbitrary autospectral density function where mean square value estimates have the same random error as the mean square value estimates of white noise with a bandwidth $B = B_s$. In other words, B_s is that value of bandwidth that makes Equations (8.46) and (8.47) correct for any Gaussian random data with an arbitrary autospectral density function of $G_{xx}(f)$.

Example 8.3. Bandwidths for Random Noise with a Nonuniform Autospectrum.

Assume a random process $\{x(t)\}$ has an autospectral density function given by

$$G_{xx}(f) = \frac{a^2}{a^2 + (f - f_r)^2} + \frac{a^2}{a^2 + (f + f_r)^2} \quad a \ll f_r$$

A plot of this autospectrum for the special case where $a = 0.1f_r$ is shown in Figure 8.2. It is seen from Figure 8.2 that the autospectrum has a single peak with a maximum value of $G_{xx}(f) \approx 1$ at the frequency $f = f_r$, and a spectral shape similar to the response of the resonant system shown in Figure 2.3 to a

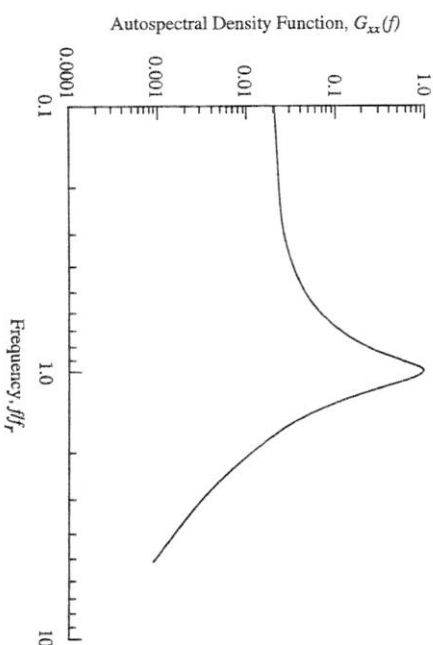


Figure 8.2 Autospectral density function of example random process.

white noise input. For this random process, the noise bandwidth B_n defined in Equation (5.106) is given by

$$B_n = \int_0^\infty \left[\frac{a^2}{a^2 + (f - f_j)^2} + \frac{a^2}{a^2 + (f + f_j)^2} \right] df = \pi a$$

while the statistical bandwidth defined in Equation (8.52) is given by

$$B_s = \frac{\left\{ \int_0^\infty \left[\frac{a^2}{a^2 + (f - f_j)^2} + \frac{a^2}{a^2 + (f + f_j)^2} \right] df \right\}^2}{\int_0^\infty \left[\frac{a^2}{a^2 + (f - f_j)^2} + \frac{a^2}{a^2 + (f + f_j)^2} \right] df} = \frac{(\pi a)^2}{\pi a / 2} = 2\pi a$$

It follows that $B_s = 2B_n$ for this particular random process. It should be mentioned that these same bandwidth descriptions can be applied to filters or spectral windows by simply substituting the squared frequency response function magnitude $|H(f)|^2$ of the filter or window for the autospectral density function $G_{yy}(f)$ in Equation (5.106) and $G_{xx}(f)$ in Equation (8.52).

8.2.3 Variance Estimates

Variance estimates can be obtained from

$$\hat{\sigma}_x^2 = \hat{\psi}_x^2 - \hat{\mu}_x^2 \quad (8.53)$$

Now

$$\text{Var}[\hat{\sigma}_x^2] = E[\hat{\sigma}_x^4] - (E[\hat{\sigma}_x^2])^2 \quad (8.54)$$

where

$$E[\hat{\sigma}_x^2] = E[\hat{\psi}_x^2] - E[\hat{\mu}_x^2] \quad (8.55)$$

$$\begin{aligned} E[\hat{\sigma}_x^4] &= E[\hat{\psi}_x^4 - 2\hat{\psi}_x^2\hat{\mu}_x^2 + \hat{\mu}_x^4] \\ &= E[\hat{\psi}_x^4] - 2E[\hat{\psi}_x^2\hat{\mu}_x^2] + E[\hat{\mu}_x^4] \end{aligned} \quad (8.56)$$

Hence

$$\text{Var}[\hat{\sigma}_x^2] = \text{Var}[\hat{\psi}_x^2] + \text{Var}[\hat{\mu}_x^2] - 2(E[\hat{\psi}_x^2\hat{\mu}_x^2] - E[\hat{\psi}_x^2]E[\hat{\mu}_x^2]) \quad (8.57)$$

Unlike mean value and mean square value estimates, variance estimates based on Equation (8.53) will be *biased* estimates. Specifically, for bandwidth-limited Gaussian white noise, substitution of Equations (8.36) and (8.30) into

Equation (8.55) shows that

$$\begin{aligned} E[\hat{\sigma}_x^2] &\approx \psi_x^2 - \left(\frac{\sigma_x^2}{2BT} + \mu_x^2 \right) \\ &\approx \sigma_x^2 \left(1 - \frac{1}{2BT} \right) \end{aligned} \quad (8.58)$$

Hence the bias error is

$$b[\hat{\sigma}_x^2] = E[\hat{\sigma}_x^2] - \sigma_x^2 \approx \frac{-\sigma_x^2}{2BT} \quad (8.59)$$

This result agrees with Equation (4.11), letting $N = 2BT$.

Consider again the important special case of bandwidth-limited Gaussian white noise where the variance terms for $\hat{\psi}_x^2$ and $\hat{\mu}_x^2$ are known from Equations (8.43) and (8.32), respectively. For Gaussian data, when $\mu_x \neq 0$, the fourth-order moment from Equation (3.78) becomes

$$E[\hat{\psi}_x^2\hat{\mu}_x^2] = E[\hat{\psi}_x^2]E[\hat{\mu}_x^2] + 2(E[\mu_x^2])^2 - 2\mu_x^4 \quad (8.60)$$

using the approximation $E[\hat{\psi}_x\mu_x] \approx E[\hat{\mu}_x^2]$, Equation (8.30) gives

$$(E[\hat{\mu}_x^2])^2 \approx \left(\frac{\sigma_x^2}{2BT} + \mu_x^2 \right)^2 \approx \frac{\mu_x^2\sigma_x^2}{BT} + \mu_x^4 \quad (8.61)$$

neglecting terms of order $(1/BT)^2$. Hence

$$E[\hat{\psi}_x^2\hat{\mu}_x^2] - E[\hat{\psi}_x^2]E[\hat{\mu}_x^2] \approx \frac{2\mu_x^2\sigma_x^2}{BT} \quad (8.62)$$

Now, substitutions of Equations (8.32), (8.43), and (8.62) into Equation (8.57) yield

$$\text{Var}[\hat{\sigma}_x^2] \approx \frac{\sigma_x^4}{BT} \quad (8.63)$$

which is independent of μ_x . The normalized random error is then given by

$$e_r[\hat{\sigma}_x^2] = \frac{\text{s.d.}[\hat{\sigma}_x^2]}{\sigma_x^2} = \frac{1}{\sqrt{BT}} \quad (8.64)$$

Note that the above result applies even if $\mu_x \neq 0$, whereas the result in Equation (8.46) for mean square value estimates should be used only when

$\mu_x = 0$. In either case, the random error can be generalized for all Gaussian random data with an arbitrary autospectral density function $G_{xx}(f)$ by substituting the statistical bandwidth B_s defined in Equation (8.52) for B in Equation (8.64). From Equation (8.10), the normalized random error for the standard deviation estimate $\hat{\sigma}_x$ is given by

$$e_r[\hat{\sigma}_x] = \frac{1}{2\sqrt{BT}} \quad (8.65)$$

The plots for the normalized random errors in mean square and rms value estimates in Figure 8.1 also apply to variance and standard deviation estimates with arbitrary mean values, as given in Equations (8.64) and (8.65).

8.3 PROBABILITY DENSITY FUNCTION ESTIMATES

Consider a probability density measurement of a single sample time-history record $x(t)$ from a stationary (ergodic) random process $\{x(t)\}$. The probability that $x(t)$ assumes values between $x - (W/2)$ and $x + (W/2)$ during a time interval T may be estimated by

$$\begin{aligned} \hat{P}[x, W] &= \text{Prob} \left[\left(x - \frac{W}{2} \right) \leq x(t) \leq \left(x + \frac{W}{2} \right) \right] \\ &= \frac{1}{T} \sum_i \Delta t_i = \frac{T_x}{T} \end{aligned} \quad (8.66)$$

where Δt_i is the time spent by $x(t)$ in this range during the i th entry into the range, and $T_x = \sum \Delta t_i$. The ratio T_x/T is the total fractional portion of the time spent by $x(t)$ in the range $[x - (W/2), x + (W/2)]$. It should be noted that T_x will usually be a function of the value x . The estimated probability $\hat{P}[x, W]$ will approach the true probability $P[x, W]$ as T approaches infinity. Moreover, this estimated probability is an unbiased estimate of the true probability. Hence

$$P[x, W] = E[\hat{P}[x, W]] = \lim_{T \rightarrow \infty} \hat{P}[x, W] = \lim_{T \rightarrow \infty} \frac{T_x}{T} \quad (8.67)$$

The probability density function $p(x)$ is defined by

$$p(x) = \lim_{W \rightarrow 0} \frac{P[x, W]}{W} = \lim_{\substack{T \rightarrow \infty \\ W \rightarrow 0}} \frac{\hat{P}[x, W]}{W} = \lim_{W \rightarrow 0} \hat{p}(x) \quad (8.68)$$

$$\hat{p}(x) = \frac{\hat{P}[x, W]}{W} = \frac{T_x}{TW} \quad (8.69)$$

where

is a sample estimate of $p(x)$. In terms of the probability density function $p(x)$, the probability of the time history $x(t)$ falling between any two values x_1 and x_2 is given by

$$\text{Prob}[x_1 \leq x(t) \leq x_2] = \int_{x_1}^{x_2} p(x) dx \quad (8.70)$$

In particular,

$$P[x, W] = \text{Prob} \left[x - \frac{W}{2} \leq x(t) \leq x + \frac{W}{2} \right] = \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (8.71)$$

Then, from Equation (8.69)

$$E[\hat{p}(x)] = \frac{E[\hat{P}[x, W]]}{W} = \frac{P[x, W]}{W} = \frac{1}{W} \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (8.72)$$

Thus for most $p(x)$,

$$E[\hat{p}(x)] \neq p(x) \quad (8.73)$$

proving that $\hat{p}(x)$ is generally a *biased* estimate of $p(x)$.

The mean square error of the estimate $\hat{p}(x)$ is calculated from Equation (8.5) by

$$E[(\hat{p}(x) - p(x))^2] = \text{Var}[\hat{p}(x)] + b^2[\hat{p}(x)] \quad (8.74)$$

where $\text{Var}[\hat{p}(x)]$ is the variance of the estimate as defined by

$$\text{Var}[\hat{p}(x)] = E[(\hat{p}(x) - E[\hat{p}(x)])^2] \quad (8.75)$$

and $b[\hat{p}(x)]$ is the bias of the estimate as defined by

$$b[\hat{p}(x)] = E[\hat{p}(x)] - p(x) \quad (8.76)$$

8.3.1 Bias of the Estimate

An expression will now be derived for the bias term of Equation (8.76). In terms of the true probability density function, Equation (8.72) shows that

$$E[\hat{p}(x)] = \frac{1}{W} \int_{x-(W/2)}^{x+(W/2)} p(\xi) d\xi \quad (8.77)$$

By expanding $p(\xi)$ in a Taylor series about the point $\xi = x$, and retaining only the first three terms,

$$p(\xi) \approx p(x) + (\xi - x)p'(x) + \frac{(\xi - x)^2}{2}p''(x) \quad (8.78)$$

From the two relations

$$\int_{x-(W/2)}^{x+(W/2)} (\xi - x) d\xi = 0 \quad (8.79)$$

and

$$\int_{x-(W/2)}^{x+(W/2)} \frac{(\xi - x)^2}{2} d\xi = \frac{W^3}{24} \quad (8.80)$$

it follows that

$$E[\hat{p}(x)] \approx p(x) + \frac{W^2}{24} p''(x) \quad (8.81)$$

Thus a first-order approximation for the bias term is given by

$$b[\hat{p}(x)] \approx \frac{W^2}{24} p''(x) \quad (8.82)$$

where $p''(x)$ is the second derivative of $p(x)$ with respect to x .

Example 8.4. Bias in Probability Density Estimate of Gaussian Random Data. Probability density estimates are generally made using a window width of $W \leq 0.2\sigma_x$. However, consider the case where a crude probability density estimate is made using a window width of $W = \sigma_x$. Assume the data being analyzed are Gaussian random noise with a mean value of zero and a variance of unity. From Equation (8.72), the expected value of the estimate is given by

$$E[\hat{p}(x)] = \frac{1}{\sqrt{2\pi}} \int_{x-0.5}^{x+0.5} e^{-x^2/2} dx$$

where solutions of the integral are available from Table A.2 in Appendix A. For example, at the mean value ($x = 0$), $E[\hat{p}(0)] = 0.3830$. From Table A.1, however, $p(0) = 0.3989$. Hence, the actual bias error in the estimate of a Gaussian probability density function at its mean value is given by Equation

(8.76) as

$$b[\hat{p}(0)] = 0.3830 - 0.3989 = -0.0159$$

where the minus sign means the estimate is less than the actual value.

Consider now the first-order approximation for the bias error given by Equation (8.82). For Gaussian data with zero mean value and unity variance,

$$p''(x) = \frac{-(1 - x^2)}{\sqrt{2\pi}} e^{-x^2/2}$$

Hence at $x = 0$, the bias is approximated by

$$b[\hat{p}(0)] = \frac{-1}{24\sqrt{2\pi}} = -0.0166$$

which is within 5% of the actual bias error computed earlier.

8.3.2 Variance of the Estimate

To evaluate the variance of an estimate $\hat{p}(x)$, it is necessary to know the statistical properties of the time intervals Δt_i that constitute T_x . Unfortunately, such time statistics for a random process are very difficult to obtain. However, the general form of an appropriate variance expression for $\hat{p}(x)$ can be established by the following heuristic argument.

From Equation (8.64), the variance of $\hat{p}(x)$ is given by

$$\text{Var}[\hat{p}(x)] = \frac{1}{W^2} \text{Var}[\hat{P}(x, W)] \quad (8.83)$$

where $\hat{P}(x, W)$ is the estimate of a proportion $P(x, W)$. The variance for a proportion estimate based on N independent sample values is given by

$$\text{Var}[\hat{P}(x, W)] = \frac{P(x, W)[1 - P(x, W)]}{N} \quad (8.84)$$

Substituting Equation (8.84) into Equation (8.83), and assuming $P(x, W) \approx Wp(x) \ll 1$, the variance of probability density estimates may be approximated by

$$\text{Var}[\hat{p}(x)] = \frac{p(x)}{NW} \quad (8.85)$$

where N is still to be determined. Now from the time-domain sampling theorem derived in Section 10.3.1, a sample record $x(t)$ of bandwidth B and length T can be completely reproduced with $N = 2BT$ discrete values. Of

course, the N discrete values will not necessarily be statistically independent. Nevertheless, for any given stationary (ergodic) random process, each sample record will represent $n = N/c^2$ independent sample values (degrees of freedom), where c is a constant. Hence from Equation (8.85),

$$\text{Var}[\hat{p}(x)] \approx \frac{c^2 p(x)}{2BTW} \quad (8.86)$$

The constant c is dependent on the autocorrelation function of the data and the sampling rate. For continuous bandwidth-limited white noise, experimental studies indicate $c \approx 0.3$. If the bandwidth-limited white noise is digitized to $N = 2BT$ discrete values, the experimental studies indicate $c = 1.0$, as would be expected from the results of Equation (8.85).

8.3.3 Normalized rms Error

The total mean square error of the probability density estimate $\hat{p}(x)$ is the sum of the variance defined in Equation (8.86) and the square of the bias defined in Equation (8.82). That is,

$$E[(\hat{p}(x) - p(x))^2] = \frac{c^2 p(x)}{2BTW} + \left[\frac{W^2 p''(x)}{24} \right]^2 \quad (8.87)$$

Hence, the normalized mean square error is approximated by

$$e^2[\hat{p}(x)] \approx \frac{c^2}{2BTWp(x)} + \frac{W^4}{576} \left[\frac{p''(x)}{p(x)} \right]^2 \quad (8.88)$$

The square root gives the normalized rms error.

It is clear from Equation (8.88) that there are conflicting requirements on the window width W in probability density measurements. On the one hand, a large value of W is desirable to reduce the random error. On the other hand, a small value of W is needed to suppress the bias error. However, the total error will approach zero as $T \rightarrow \infty$ if W is restricted so that $W \rightarrow 0$ and $WT \rightarrow \infty$. In practice, values of $W \leq 0.2\sigma_x$ will usually limit the normalized bias error to less than 1%. This is true because of the $p''(x)$ term in the bias portion of the error given by Equation (8.88). Probability density functions of common (approximately Gaussian) random data do not display abrupt or sharp peaks, which are indicative of a large second derivative.

8.3.4 Joint Probability Density Function Estimates

Joint probability density function estimates for a pair of sample time-history records $x(t)$ and $y(t)$ from two stationary (ergodic) random processes $\{x(t)\}$

and $\{y(t)\}$ may be defined as follows. Analogous to Equation (8.66), let

$$\hat{P}[x, W_x; y, W_y] = \frac{T_{x,y}}{T} \quad (8.89)$$

estimate the joint probability that $x(t)$ is inside the interval W_x centered at x while simultaneously $y(t)$ is inside the interval W_y centered at y . This is measured by the ratio $T_{x,y}/T$, where $T_{x,y}$ represents the amount of time that these two events coincide in time T . Clearly, $T_{x,y}$ will usually be a function of both x and y . This estimated joint probability will approach the true probability $P[x, W_x; y, W_y]$ as T approaches infinity, namely,

$$P[x, W_x; y, W_y] = \lim_{T \rightarrow \infty} \hat{P}[x, W_x; y, W_y] = \lim_{T \rightarrow \infty} \frac{T_{x,y}}{T} \quad (8.90)$$

The joint probability density function $p(x, y)$ is now defined by

$$p(x, y) = \lim_{\substack{W_x \rightarrow 0 \\ W_y \rightarrow 0}} \frac{P[x, W_x; y, W_y]}{W_x W_y} = \lim_{\substack{T \rightarrow \infty \\ W_x \rightarrow 0 \\ W_y \rightarrow 0}} \frac{\hat{P}[x, W_x; y, W_y]}{W_x W_y} = \lim_{\substack{T \rightarrow \infty \\ W_x \rightarrow 0 \\ W_y \rightarrow 0}} \hat{p}(x, y) \quad (8.91)$$

where

$$\hat{p}(x, y) = \frac{\hat{P}[x, W_x; y, W_y]}{W_x W_y} = \frac{T_{x,y}}{TW_x W_y} \quad (8.92)$$

Assume that W_x and W_y are sufficiently small that the bias errors are negligible. Then the mean square error associated with the estimate $\hat{p}(x, y)$ will be given by the variance of the estimate. As for first-order probability density estimates, this quantity is difficult to determine precisely by theoretical arguments alone. However, by using the same heuristic arguments that produced Equation (8.86), a general form for the variance can be approximated. Specifically, for the special case where $x(t)$ and $y(t)$ are both bandwidth limited white noise with identical bandwidths B ,

$$\text{Var}[\hat{p}(x, y)] \approx \frac{c^2 p(x, y)}{2BTW_x W_y} \quad (8.93)$$

where c is an unknown constant.

8.4 CORRELATION FUNCTION ESTIMATES

Consider now two sample time-history records $x(t)$ and $y(t)$ from two stationary (ergodic) random processes $\{x(t)\}$ and $\{y(t)\}$. The next statistical quantities of interest are the stationary autocorrelation functions $R_{xx}(\tau)$ and $R_{yy}(\tau)$ and

the cross-correlation function $R_{xy}(\tau)$. To simplify the following derivation, the mean values μ_x and μ_y will be assumed to be zero. For continuous data, $x(t)$ and $y(t)$, which exist only over a time interval T , the sample cross-correlation estimate $\hat{R}_{xy}(\tau)$ can be defined by

$$\hat{R}_{xy}(\tau) = \begin{cases} \frac{1}{T-\tau} \int_0^{T-\tau} x(t)y(t+\tau) dt & 0 \leq \tau < T \\ \frac{1}{T-|\tau|} \int_{|\tau|}^T x(t)y(t+\tau) dt & -T < \tau \leq 0 \end{cases} \quad (8.94)$$

To avoid the use of absolute value signs, τ will be considered positive henceforth since a similar proof applies for negative τ . The sample autocorrelation function estimates $\hat{R}_{xx}(\tau)$ and $\hat{R}_{yy}(\tau)$ are merely special cases when the two records coincide. That is,

$$\begin{aligned} \hat{R}_{xx}(\tau) &= \frac{1}{T-\tau} \int_0^{T-\tau} x(t)x(t+\tau) dt & 0 \leq \tau < T \\ \hat{R}_{yy}(\tau) &= \frac{1}{T-\tau} \int_0^{T-\tau} y(t)y(t+\tau) dt & 0 \leq \tau < T \end{aligned} \quad (8.95)$$

Thus by analyzing the cross-correlation function estimate, one derives results that are also applicable to the autocorrelation function estimates.

If the data exist for time $T + \tau$ instead of only for time T , then an alternative definition for $\hat{R}_{xy}(\tau)$ is

$$\hat{R}_{xy}(\tau) = \frac{1}{T} \int_0^T x(t)y(t+\tau) dt \quad 0 \leq \tau < T \quad (8.96)$$

This formula has a fixed integration time T instead of a variable integration time as in Equation (8.94), and is the way the correlation functions have been defined previously. Note that for either Equation (8.94) or Equation (8.96), mean square estimates of $x(t)$ or $y(t)$ are merely special cases when $\tau = 0$. For simplicity in notation, Equation (8.96) will be used in the following development instead of Equation (8.94). The same final results are obtained for both definitions, assuming the data exist for time $T + \tau$.

The expected value of the estimate $\hat{R}_{xy}(\tau)$ is given by

$$\begin{aligned} E[\hat{R}_{xy}(\tau)] &= \frac{1}{T} \int_0^T E[x(t)y(t+\tau)] dt \\ &= \frac{1}{T} \int_0^T R_{xy}(\tau) dt = R_{xy}(\tau) \end{aligned} \quad (8.97)$$

Hence $\hat{R}_{xy}(\tau)$ is an unbiased estimate of $R_{xy}(\tau)$, independent of T . The mean

square error is given by the variance

$$\begin{aligned} \text{Var}[\hat{R}_{xy}(\tau)] &= E[(\hat{R}_{xy}(\tau) - R_{xy}(\tau))^2] = E[\hat{R}_{xy}^2(\tau) - R_{xy}^2(\tau)] \\ &= \frac{1}{T^2} \int_0^T \int_0^T (E[x(u)y(u+\tau)x(v)y(v-\tau)] - R_{xy}^2(\tau)) dv du \end{aligned} \quad (8.98)$$

At this point, in order to simplify the later mathematical analysis and to agree with many physical cases of interest, it will be assumed that the random processes $\{x(t)\}$ and $\{y(t)\}$ are jointly Gaussian for any set of fixed times. This restriction may be removed by substituting certain integrability conditions on the non-Gaussian parts of the random processes without altering in any essential way the results to be derived. When $\{x(t)\}$ and $\{y(t)\}$ are jointly Gaussian, it follows that $\{x(t)\}$ and $\{y(t)\}$ are separately Gaussian.

For Gaussian stationary random processes with zero mean values, the fourth-order statistical expression is obtained from Equation (5.135) as follows:

$$\begin{aligned} E[x(u)y(u+\tau)x(v)y(v-\tau)] &= R_{xy}^2(\tau) + R_{xx}(v-u)R_{yy}(v-u) \\ &\quad + R_{xy}(v-u+\tau)R_{yx}(v-u-\tau) \end{aligned} \quad (8.99)$$

Hence the variance expression may be written as

$$\begin{aligned} \text{Var}[\hat{R}_{xy}(\tau)] &= \frac{1}{T^2} \int_0^T \int_0^T (R_{xx}(v-u)R_{yy}(v-u) \\ &\quad + R_{xy}(v-u+\tau)R_{yx}(v-u-\tau)) dv du \\ &\quad + \frac{1}{T} \int_{-\tau}^T \left(1 - \frac{|\xi|}{T}\right) (R_{xx}(\xi)R_{yy}(\xi) + R_{xy}(\xi+\tau)R_{yx}(\xi-\tau)) d\xi \end{aligned} \quad (8.100)$$

The second expression occurs from the first by letting $\xi = v - u$, $d\xi = dv$, and then reversing the order of integration between ξ and u . Now,

$$\lim_{T \rightarrow \infty} T \text{Var}[\hat{R}_{xy}(\tau)] = \int_{-\infty}^{\infty} (R_{xx}(\xi)R_{yy}(\xi) + R_{xy}(\xi+\tau)R_{yx}(\xi-\tau)) d\xi < \infty \quad (8.101)$$

assuming $R_{xx}(\xi)$, $R_{yy}(\xi)$ and $R_{xy}(\xi)$, $R_{yx}(\xi)$ are absolutely integrable over $(-\infty, \infty)$. This proves that $\hat{R}_{xy}(\tau)$ is a consistent estimate of $R_{xy}(\tau)$, and for large T , has a variance given by

$$\text{Var}[\hat{R}_{xy}(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} (R_{xx}(\xi)R_{yy}(\xi) + R_{xy}(\xi+\tau)R_{yx}(\xi-\tau)) d\xi \quad (8.102)$$

Several special cases of Equation (8.102) are worthy of note. For autocorrelation estimates, Equation (8.102) becomes

$$\text{Var}[\hat{R}_{xx}(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} (R_{xx}^2(\xi) + R_{xx}(\xi + \tau)R_{xx}(\xi - \tau)) d\xi \quad (8.103)$$

At the zero displacement point $\tau = 0$.

$$\text{Var}[\hat{R}_{xx}(0)] \approx \frac{2}{T} \int_{-\infty}^{\infty} R_{xx}^2(\xi) d\xi \quad (8.104)$$

The assumption that $R_{xx}(\tau)$ approaches zero for large τ shows that

$$R_{xx}^2(\xi) \gg R_{xx}(\xi + \tau)R_{xx}(\xi - \tau) \quad \text{for } \tau \gg 0$$

Hence for large τ ,

$$\text{Var}[\hat{R}_{xx}(\tau)] \approx \frac{1}{T} \int_{-\infty}^{\infty} R_{xx}^2(\xi) d\xi \quad (8.105)$$

which is one-half the value of Equation (8.104).

8.4.1 Bandwidth-Limited Gaussian White Noise

Consider the special case where $\{x(t)\}$ is bandwidth-limited Gaussian white noise with a mean value $\mu_x = 0$ and a bandwidth B as defined in Equations (8.25) and (8.26). For sample records $x(t)$ of length T , from Equation (8.103), the variance for $\hat{R}_{xx}(\tau)$ is given conservatively by

$$\text{Var}[\hat{R}_{xx}(\tau)] \approx \frac{1}{2BT} [R_{xx}^2(0) + R_{xx}^2(\tau)] \quad (8.106)$$

This reduces to Equation (8.45) at the point $\tau = 0$. Similarly, when both $x(t)$ and $y(t)$ are samples of length T from bandwidth-limited Gaussian white noise with mean values $\mu_x = \mu_y = 0$ and identical bandwidths B , it follows from Equation (8.102) that

$$\text{Var}[\hat{R}_{xy}(\tau)] \approx \frac{1}{2BT} [R_{xx}(0)R_{yy}(0) + R_{xy}^2(\tau)] \quad (8.107)$$

This result requires that T be sufficiently large that Equation (8.102) can replace Equation (8.100). Satisfactory conditions in practice are $T \geq 10|\tau|$ and $BT \geq 5$.

For $\mu_x = \mu_y = 0$ and $R_{xy}(\tau) \neq 0$, the normalized mean square error is given by

$$\varepsilon^2[\hat{R}_{xy}(\tau)] = \frac{\text{Var}[\hat{R}_{xy}(\tau)]}{R_{xy}^2(\tau)} \approx \frac{1}{2BT} \left[1 + \frac{R_{xx}(0)R_{yy}(0)}{R_{xy}^2(\tau)} \right] \quad (8.108)$$

The square root gives the normalized rms error ε , which includes only a random error term because the bias error is zero if the record length is longer than $T + \tau$. Thus, for cross-correlation function estimates,

$$\varepsilon[\hat{R}_{xy}(\tau)] \approx \frac{1}{\sqrt{2BT}} [1 + \rho_{xy}^2(\tau)]^{1/2} \quad (8.109)$$

where

$$\rho_{xy}(\tau) = \frac{R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \quad (8.110)$$

is the cross-correlation coefficient function. Note that $R_{xy}(\tau) = C_{xy}(\tau)$ when $\mu_x = \mu_y = 0$. When $x(t) = y(t)$, at the special point $\tau = 0$, the quantity $R_{xx}(0) = \psi_x^2$ and $\rho_{xx}(0) = 1$ so that one obtains

$$\varepsilon[\hat{R}_{xx}(0)] = \varepsilon[\hat{\psi}_x^2] \approx \frac{1}{\sqrt{BT}} \quad (8.111)$$

in agreement with Equation (8.46). In general, for autocorrelation function estimates,

$$\varepsilon[\hat{R}_{xx}(\tau)] \approx \frac{1}{\sqrt{2BT}} [1 + \rho_{xx}^2(\tau)]^{1/2} \quad (8.112)$$

where

$$\rho_{xx}(\tau) = \frac{R_{xx}(\tau)}{R_{xx}(0)} \quad (8.113)$$

Equation (8.112) is plotted in Figure 8.3.

8.4.2 Noise-to-Signal Considerations

Some applications of Equation (8.109) are worthy of note. Suppose $x(t)$ and $y(t)$ are given by

$$\begin{aligned} x(t) &= s(t) + m(t) \\ y(t) &= s(t) + n(t) \end{aligned} \quad (8.114)$$

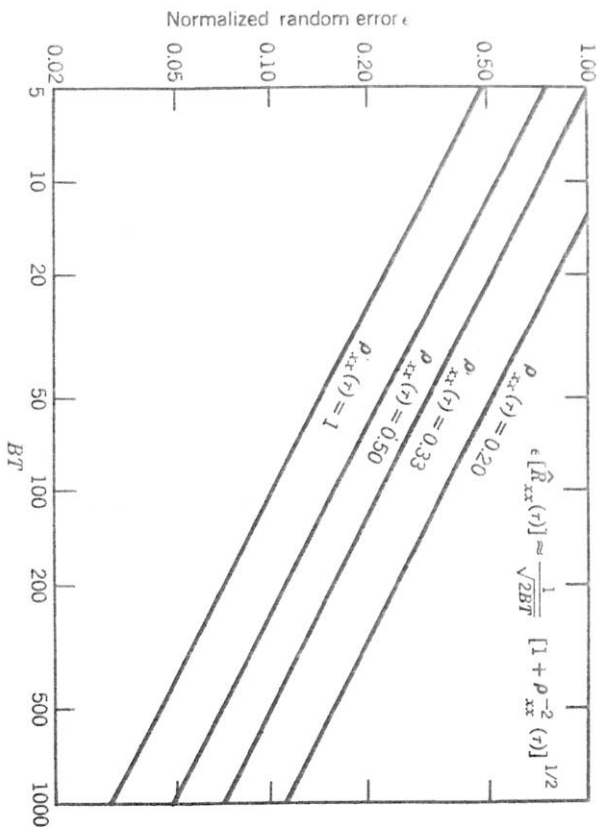


Figure 8.3 Normalized random error of autocorrelation function estimates.

where $s(t)$, $m(t)$, and $n(t)$ are mutually uncorrelated. Then

$$\begin{aligned} R_{xy}(\tau) &= R_{ss}(\tau) = R_{ss}(0)\rho_{ss}(\tau) = S\rho_{ss}(\tau) \\ R_{ss}(0) &= R_{ss}(0) + R_{mm}(0) = S + M \\ R_{yy}(0) &= R_{ss}(0) + R_{nn}(0) = S + N \end{aligned} \quad (8.115)$$

It follows from Equation (8.109) that the normalized mean square error is

$$e^2[\hat{R}_{xy}(\tau)] \approx \frac{1}{2BT} \left[1 + \frac{(S + M)(S + N)}{S^2 \rho_{ss}^2(\tau)} \right] \quad (8.116)$$

At $\tau = 0$, where $\rho_{ss}(0) = 1$,

$$e^2[\hat{R}_{xy}(0)] \approx \frac{1}{2BT} [2 + (M/S) + (N/S) + (M/S)(N/S)] \quad (8.117)$$

These relations are useful for two-detector systems where $x(t)$ and $y(t)$ of Equation (8.114) measure the outputs of the two detectors containing a common signal $s(t)$ but uncorrelated noises $m(t)$ and $n(t)$.

8.4 CORRELATION FUNCTION ESTIMATES

297

CASE 1. $M = 0$ where $(N/S) \gg 1$:

$$e^2[\hat{R}_{xy}(0)] \approx \frac{1}{2BT} \left(\frac{N}{S} \right) \quad (8.118)$$

CASE 2. $M = N$ where $(N/S) \gg 1$:

$$e^2[\hat{R}_{xy}(0)] \approx \frac{1}{2BT} \left(\frac{N}{S} \right)^2 \quad (8.119)$$

These two cases can represent important physical applications. For example, in Case 1, $x(t)$ might be a noise-free reference signal being cross-correlated against a corrupted received signal $y(t)$. In Case 2, a corrupted received signal $y(t)$ is being autocorrelated against itself. For any given value of $(N/S) \gg 1$, a much larger value of BT is required in Case 2 than in Case 1 to achieve a desired mean square error. Applications for these and other formulas in this chapter are given in Reference 8.1.

8.4.3 Location Estimates of Peak Correlation Values

Previous results in this section give random error formulas that indicate how well one can estimate the magnitudes of peak values of $R_{xy}(\tau)$ and $R_{xx}(\tau)$. There remains the difficult related problem of determining the precise location where these peak correlation values occur. For definiteness, assume that $R_{xx}(\tau)$ has the form associated with bandwidth limited white noise, namely,

$$R_{xx}(\tau) = R_{xx}(0) \left(\frac{\sin 2\pi B\tau}{2\pi B\tau} \right) \quad (8.120)$$

The maximum value of $R_{xx}(\tau)$ occurs at $\tau = 0$. Expansion of $\sin 2\pi B\tau$ near $\tau = 0$ yields

$$\sin 2\pi B\tau \approx (2\pi B\tau) - \frac{(2\pi B\tau)^3}{6} \quad (8.121)$$

Thus near $\tau = 0$, the estimate $\hat{R}_{xx}(\tau)$ is given by

$$\hat{R}_{xx}(\tau) \approx \left[1 - \frac{2(\pi B\tau)^2}{3} \right] R_{xx}(0) \quad (8.122)$$

where the mean value is

$$E[\hat{R}_{xx}(0)] = R_{xx}(0) \quad (8.123)$$

Hence $\hat{R}_{xx}(0)$ is an unbiased estimate of $R_{xx}(0)$. The variance in these estimates is given by

$$\text{Var}[\hat{R}_{xx}(0)] = E[(\hat{R}_{xx}(0) - R_{xx}(0))^2] \approx \frac{4}{9}(\pi B)^4 R_{xx}^2(0) E[\tau^4] \quad (8.124)$$

The normalized mean square error is then

$$e^2[\hat{R}_{xx}(0)] = \frac{\text{Var}[\hat{R}_{xx}(0)]}{R_{xx}^2(0)} \approx \frac{4}{9}(\pi B)^4 E[\tau^4] \quad (8.125)$$

Assume next that these values of τ are such that τ follows a Gaussian distribution with zero mean value and variance $\sigma_1^2(\tau)$, namely,

$$\mu_1(\tau) = E[\tau] = 0 \quad (8.126)$$

$$\sigma_1^2(\tau) = E[\tau^2]$$

Then, the fourth-order moment in Equation (8.125) satisfies

$$E[\tau^4] = 3\sigma_1^4(\tau) \quad (8.127)$$

It follows that

$$e^2[\hat{R}_{xx}(0)] \approx \frac{4}{9}(\pi B)^4 \sigma_1^4(\tau) \quad (8.128)$$

This proves that

$$\sigma_1(\tau) \approx \frac{0.93}{\pi B} \{e[\hat{R}_{xx}(0)]\}^{1/2} \quad (8.129)$$

The 95% confidence interval for determining the location where the peak value occurs is now

$$[-2\sigma_1(\tau) \leq \tau \leq 2\sigma_1(\tau)] \quad (8.130)$$

Example 8.5. Time-Delay Estimate from Cross-Correlation Calculation.

Assume two received time-history records, $x(t)$ and $y(t)$, contain a common signal $s(t)$ and uncorrelated noises, $m(t)$ and $n(t)$, such that

$$x(t) = s(t) + m(t)$$

$$y(t) = s(t + \tau_0) + n(t)$$

From the developments in Section 5.1.4, the maximum value in the cross-correlation function

$$R_{xy}(\tau) = R_{sx}(\tau - \tau_0)$$

will occur at $\tau = \tau_0$, which defines the time delay between the received signal $s(t)$ in $x(t)$ and $y(t)$. Now assume $s(t)$ represents bandwidth-limited white noise with a bandwidth of $B = 100$ Hz, and the noise-to-signal ratios in $x(t)$ and $y(t)$ are given by $M/S = N/S = 10$. If the available record lengths for $x(t)$ and $y(t)$ are $T = 5$ sec, determine the accuracy of the time-delay estimate τ_0 based on the time of the maximum value of the cross-correlation estimate $\hat{R}_{xy}(\tau)$.

From Equation (8.119), the normalized random error of the cross-correlation estimate at its maximum value is approximated by

$$e[\hat{R}_{xy}(\tau_0)] \approx e[\hat{R}_{ss}(0)] \approx \left[\frac{10^2}{(2)(100)(5)} \right]^{1/2} = 0.32$$

Note that a more exact error is given by Equation (8.117) as $e = 0.35$. Now the standard deviation of the estimate $\hat{\tau}_0$ is approximated from Equation (8.129) by

$$\sigma_1(\tau) \approx \frac{0.93(0.35)^{1/2}}{\pi(100)} = 0.0017 \text{ sec}$$

Hence, from Equation (8.130), an approximate 95% confidence interval for the time delay τ_0 in seconds is given by

$$[\hat{\tau}_0 - 0.0034 \leq \tau_0 \leq \hat{\tau}_0 + 0.0034]$$

It should be mentioned that more accurate time delay estimates can often be achieved using the Hilbert transform procedures discussed in Chapter 13 or the phase analysis procedures detailed in Reference 8.2.

8.5 AUTOSPECTRAL DENSITY FUNCTION ESTIMATES

Assume the autospectral density function associated with a sample time-history record $x(t)$ is estimated using the filtering-squaring-averaging procedure detailed in Section 5.2.3 and illustrated in Figure 5.5. From Equation (5.68), the autospectral density function estimate computed by this procedure is

$$\hat{G}_{xx}(f) = \frac{1}{B_e T} \int_0^T x^2(f, B_e, t) dt = \frac{\hat{\psi}_x^2(f, B_e)}{B_e} \quad (8.131)$$

where $x(f, B_e, t)$ is that portion of $x(t)$ passed by an ideal rectangular bandpass filter with a bandwidth of B_e centered at frequency f , and $\hat{\psi}_x^2(f, B_e)$ is an estimate of the mean square value of $x(f, B_e, t)$. The true mean square value that would be computed with an infinite averaging time is given by

$$\psi_x^2(f, B_e) = \int_{f-(B_e/2)}^{f+(B_e/2)} G_{xx}(\xi) d\xi \quad (8.132)$$

Equations (8.131) and (8.132) show that

$$E[\hat{G}_{xx}(f)] = \frac{\psi_x(f, B_e)}{B_e} = \frac{1}{B_e} \int_{f-(B_e/2)}^{f+(B_e/2)} G_{xx}(\xi) d\xi \quad (8.133)$$

Thus, for most $G_{xx}(f)$,

$$E[\hat{G}_{xx}(f)] \neq G_{xx}(f) \quad (8.134)$$

so that $\hat{G}_{xx}(f)$ is generally a *biased* estimate of $G_{xx}(f)$.

The mean square error of the estimate $\hat{G}_{xx}(f)$ is calculated from Equation (8.5) by

$$E[(\hat{G}_{xx}(f) - G_{xx}(f))^2] = \text{Var}[\hat{G}_{xx}(f)] + b^2[\hat{G}_{xx}(f)] \quad (8.135)$$

where $\text{Var}[\hat{G}_{xx}(f)]$ is the variance of the estimate as defined by

$$\text{Var}[\hat{G}_{xx}(f)] = E[(\hat{G}_{xx}(f) - E[\hat{G}_{xx}(f)])^2] \quad (8.136)$$

and $b[\hat{G}_{xx}(f)]$ is the bias of the estimate as defined by

$$b[\hat{G}_{xx}(f)] = E[\hat{G}_{xx}(f)] - G_{xx}(f) \quad (8.137)$$

8.5.1 Bias of the Estimate

An expression for the bias term of Equation (8.136) may be derived by the procedure used to derive the bias term for probability density estimates in Section 8.3.1. Specifically, by expanding $G_{xx}(\xi)$ in a Taylor series about the point $\xi = f$ and retaining only the first three terms, it follows from Equation (8.133) that

$$E[\hat{G}_{xx}(f)] \approx G_{xx}(f) + \frac{B_e^2}{24} G_{xx}''(f) \quad (8.138)$$

Thus the bias term is approximated by

$$b[\hat{G}_{xx}(f)] \approx \frac{B_e^2}{24} G_{xx}''(f) \quad (8.139)$$

where $G_{xx}''(f)$ is the second derivative of $G_{xx}(f)$ with respect to f , and is related to $R_{xx}(\tau)$ by the expression

$$G_{xx}''(f) = -8\pi^2 \int_{-\infty}^{\infty} \tau^2 R_{xx}(\tau) e^{-j2\pi f\tau} d\tau \quad (8.140)$$

It should be emphasized that Equation (8.139) is only a first-order approximation of the bias error, which is applicable for cases where $B_e^2 G_{xx}''(f) < G_{xx}(f)$. Because autospectra in practice often display sharp peaks reflecting large second derivatives, this may provide an inadequate measure of the bias error in some cases. Generally speaking, Equation (8.139) will exaggerate the degree of bias in estimates where $B_e^2 G_{xx}''(f)$ is large.

The bias error in Equation (8.139) is derived assuming the spectral estimation is accomplished using an ideal rectangular spectral window as defined in Equation (8.132) and illustrated in Figure 8.4. It will be seen in Section 11.5 that spectral estimation procedures in practice involve spectral windows that deviate widely from an ideal rectangular form. Nevertheless, Equation (8.139) constitutes a useful first-order approximation that correctly describes important qualitative results. In particular, the bias error increases as $G''(f)$ increases for a given B_e , or as B_e increases for a given $G''(f)$. Also, it is clear from Figure 8.4 that the bias error is always in the direction of reduced dynamic range; that is, spectral density peaks are underestimated and spectral density valleys are overestimated.

Example 8.6. Illustration of Bias Error in Autospectrum Estimate. Assume white noise is applied to a single degree-of-freedom system defined in Section 2.4.1. From Example 6.3, the displacement response of the system will have an autospectral density function given by

$$G_y(f) = \frac{G}{[1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2}$$

where G is the spectral density of the excitation in displacement units, f_n is the undamped natural frequency of the system, and ζ is the damping ratio of the

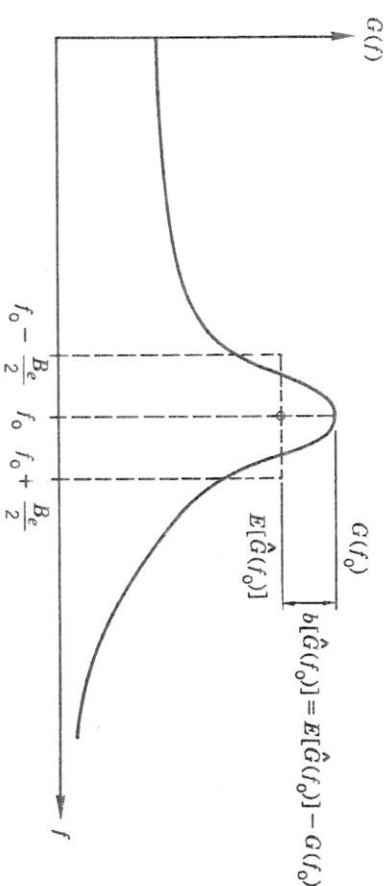


Figure 8.4 Illustration of bias error introduced by frequency smoothing of spectral density estimates.

system. As pointed out in Equation (2.25), the peak value of $G_{yy}(f)$ occurs at the resonance frequency $f_r = f_n \sqrt{1 - 2\zeta^2}$. For the case where $\zeta \ll 1$ so that $f_r \approx f_n$, the second derivative of $G_{yy}(f)$ at the frequency f_r yields

$$G''_{yy}(f_r) \approx (-8/B_r^2)G_{yy}(f_r)$$

where B_r is the half-power point bandwidth of the resonance peak given by Equation (2.27) as

$$B_r \approx 2\zeta f_r$$

Hence, the normalized bias error from Equation (8.139) becomes

$$e_b[\hat{G}_{yy}(f_r)] = \frac{b[\hat{G}_{yy}(f_r)]}{G_{yy}(f_r)} = -\frac{1}{3} \left(\frac{B_e}{B_r} \right)^2 \quad (8.141)$$

This concludes Example 8.6.

To check the accuracy of Equation (8.141), consider a more exact solution for the bias error in the autospectral density analysis of the response of a single degree-of-freedom system to white noise when $\zeta \ll 1$ so that $f_r \approx f_n$. Specifically, from Equation (8.133),

$$E[\hat{G}_{yy}(f)] = \frac{1}{B_e} \int_{f_r - (B_e/2)}^{f_r + (B_e/2)} G_{yy}(f) df = \frac{G}{B_e} \int_{f_r - (B_e/2)}^{f_r + (B_e/2)} \frac{df}{[1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2}$$

Now

$$\begin{aligned} [1 - (f/f_n)^2]^2 + [2\zeta f/f_n]^2 &= \{[(f/f_n) - 1][(f/f_n) + 1]\}^2 + [2\zeta f/f_n]^2 \\ &\approx 4[(f/f_n) - 1]^2 + 4\zeta^2 \quad \text{when } f \approx f_n \end{aligned}$$

Hence,

$$E[\hat{G}_{yy}(f)] = \frac{G}{4B_e} \int_{f_r - (B_e/2)}^{f_r + (B_e/2)} \frac{df}{[(f/f_n) - 1]^2 + \zeta^2}$$

Letting $x = (f/f_n) - 1$, it follows that $dx = df/f_n$ and

$$E[\hat{G}_{yy}(f)] = \frac{Gf_n}{4B_e} \int_{-B_e/2f_n}^{B_e/2f_n} \frac{dx}{x^2 + \zeta^2}$$

From integral tables,

$$\int \frac{dx}{x^2 + \zeta^2} = \frac{1}{\zeta} \tan^{-1} \left(\frac{x}{\zeta} \right)$$

Hence

$$E[\hat{G}_{yy}(f)] = \frac{Gf_n}{4\zeta B_e} \left[\tan^{-1} \left(\frac{B_e}{2\zeta f_n} \right) - \tan^{-1} \left(\frac{-B_e}{2\zeta f_n} \right) \right] = \frac{Gf_n}{2\zeta B_e} \tan^{-1} \left(\frac{B_e}{2\zeta f_n} \right)$$

Noting that the half-power bandwidth $B_r \approx 2\zeta f_r \approx 2\zeta f_n$ when $\zeta \ll 1$, it follows that

$$E[\hat{G}_{yy}(f)] = \frac{Gf_n}{2\zeta B_e} \tan^{-1} \left(\frac{B_e}{B_r} \right)$$

At $f = f_n$, the actual value of $G_{yy}(f)$ is

$$G_{yy}(f_n) = \frac{G}{4\zeta^2} = G_{yy}(f)$$

Thus,

$$\frac{E[\hat{G}_{yy}(f)]}{G_{yy}(f)} = \frac{B_r}{B_e} \tan^{-1} \left(\frac{B_e}{B_r} \right)$$

which is in agreement with a formula determined by different procedures in Reference 8.3. In terms of the normalized bias error defined in Equation (8.9b),

$$e_b[\hat{G}_{yy}(f)] = \frac{B_r}{B_e} \tan^{-1} \left(\frac{B_e}{B_r} \right) - 1 \quad (8.142)$$

The normalized bias error results given by Equations (8.141) and (8.142) are compared in Figure 8.5. It is seen from this figure that the two results are in good agreement for resolution to half-power point bandwidth ratios of $B_e/B_r \leq 0.5$, which corresponds to a normalized bias error of about -9% . When adequate record lengths are available, it is common practice to choose $B_e = B_r/4$ so that a negligible bias error of $e_b \approx -2\%$ is obtained. Even for relatively short record lengths where the resolution bandwidth might be selected using the minimum mean square error criterion in Section 8.5.3 to follow, it is unusual to choose a resolution bandwidth of greater than $B_e = B_r/2$. Hence, the simpler and more conservative result in Equation (8.141) is usually an acceptable approximation for the maximum bias error in autospectral density estimates of data representing the response of lightly

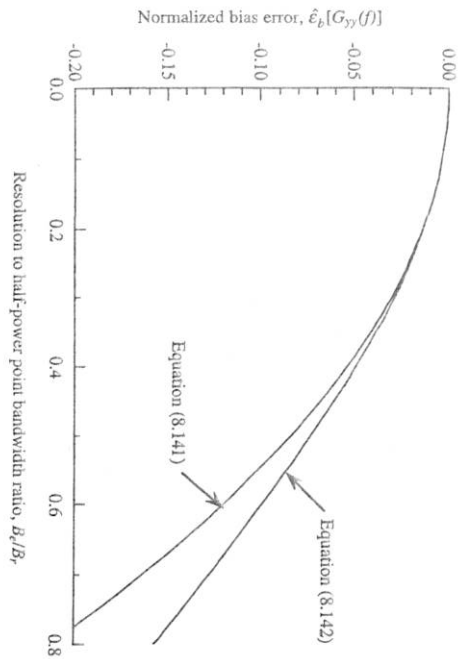


Figure 8.5 Normalized bias error for autospectral density estimates of single degree-of-freedom system response to white noise.

damped mechanical and electrical systems. See Reference 8.4 for further details on the bias errors in autospectral density analysis when the resolution filter has other than an ideal rectangular bandwidth.

8.5.2 Variance of the Estimate

The most direct way to arrive at a variance expression for autospectral density estimates is to apply the results derived in Section 8.2.2, as follows. From Equation (8.131), the estimate

$$B_e \hat{G}_{xx}(f) = \hat{\psi}_x^2(f, B_e) \quad (8.143)$$

is an unbiased estimate of the mean square value of $x(t)$ within the bandwidth B_e centered at f . The true value is given by $\psi_x^2(f, B_e) = B_e G_{xx}(f)$ when $G_{xx}(f)$ is constant over the bandwidth B_e . This will be approximately the case if B_e is sufficiently small. The result of Equation (8.45) applies to these estimates with $R_{xx}(0) = B_e G_{xx}(f)$. Hence

$$\text{Var}[B_e \hat{G}_{xx}(f)] \approx \frac{B_e^2 G_{xx}^2(f)}{B_e T} \quad (8.144)$$

But, because B_e is a constant,

$$\text{Var}[B_e \hat{G}_{xx}(f)] = B_e^2 \text{Var}[\hat{G}_{xx}(f)] \quad (8.145)$$

This gives for the variance of the estimate

$$\text{Var}[\hat{G}_{xx}(f)] \approx \frac{G_{xx}^2(f)}{B_e T} \quad (8.146)$$

The result in Equation (8.146) is based on the assumption that the filtered data behave like bandwidth-limited Gaussian white noise. This is an excellent assumption in practice when the filter resolution bandwidth B_e is small. The central limit theorem applies to indicate that the filtered data should be more Gaussian than the input data, and the fact that B_e is small means that the output spectrum must be essentially constant. Hence, for small B_e , one can strongly state that the normalized random error will be

$$\epsilon_r[\hat{G}_{xx}(f)] \approx \frac{1}{\sqrt{B_e T}} \quad (8.147)$$

Note that this result is independent of frequency.

8.5.3 Normalized rms Error

The total mean square error of the autospectral density estimate $\hat{G}_{xx}(f)$ is the sum of the variance defined in Equation (8.146) and the square of the bias defined in Equation (8.139). That is,

$$E[(\hat{G}_{xx}(f) - G_{xx}(f))^2] \approx \frac{G_{xx}^2(f)}{B_e T} + \left[\frac{B_e^2 G_{xx}''(f)}{24} \right]^2 \quad (8.148)$$

Hence, the normalized mean square error is approximated by

$$\epsilon^2[\hat{G}_{xx}(f)] = \frac{E[(\hat{G}_{xx}(f) - G_{xx}(f))^2]}{G_{xx}^2(f)} \approx \frac{1}{B_e T} + \frac{B_e^4}{576} \left[\frac{G_{xx}''(f)}{G_{xx}(f)} \right]^2 \quad (8.149)$$

The square root gives the normalized rms error.

Two important features of the error expression for autospectral density estimates should be noted. First, there are conflicting requirements on the resolution bandwidth B_e , namely, a small value of B_e is needed to suppress the bias portion of the error while a large value of B_e is desired to reduce the random portion of the error. This is similar to the situation discussed in Section 8.3.3 for the window width W in probability density measurements. The problem here, however, is more critical since autospectra in practice often display sharp peaks (large second derivatives), which aggravate the bias error problem. Second, the random portion of the error includes only the resolution bandwidth B_e , and not the total data bandwidth B . Hence the random portion of the error is a function primarily of analysis parameters rather than unknown

data parameters. This greatly enhances the practical value of Equation (8.149) in experimental design and data analysis.

Example 8.7. Illustration of Optimum Resolution Bandwidth Selection. As in Example 8.6, consider the response of a single degree-of-freedom system to white noise. The maximum frequency resolution bias error in an autospectral density estimate of the response data for this case is given by Equation (8.141). Hence, the maximum normalized mean square error in an autospectral density estimate is given from Equation (8.149) by

$$\varepsilon^2[\hat{G}_{xx}(f)] \approx \frac{1}{B_e T} + \frac{1}{9} \left(\frac{B_e}{B_r} \right)^4$$

If one assumes that the system has light damping ($\zeta \ll 1$), then Equation (2.27) applies and

$$\varepsilon^2[\hat{G}_{xx}(f)] \approx \frac{1}{B_e T} + \frac{1}{9} \left(\frac{B_e}{2\zeta^2 f_r} \right)^4$$

The resolution bandwidth that will minimize the normalized mean square error is obtained by taking the derivative of the error with respect to B_e , equating to zero, and solving for the *optimum resolution bandwidth* B_o . This yields

$$B_o \approx 2 \frac{(\zeta f_r)^{4/5}}{T^{1/5}} \quad (8.150)$$

See Reference 8.5 for further discussions of optimum resolution bandwidths for the computation of autospectral density estimates.

Two aspects of Equation (8.150) are important to note. First, the optimum resolution bandwidth is inversely proportional to the one-fifth power of the averaging time T , meaning that the optimum bandwidth is not very sensitive to the averaging time. For example, the value of B_o for $T = 1$ sec is less than 60% greater than the B_o for $T = 10$. Second, the optimum resolution bandwidth is a function of the four-fifths power of the system resonant frequency f_r , meaning that the optimum bandwidth is almost directly proportional to the resonance frequency assuming a constant damping ratio ζ . This explains why a resolution bandwidth that is proportional to the bandwidth center frequency (referred to as a *constant percentage resolution bandwidth*) is often recommended for the autospectral density analysis of random data representing the resonant response of lightly damped mechanical and electrical systems.

8.5.4 Estimates from Finite Fourier Transforms

In Section 8.5.2, an expression is derived for the variance of autospectral density estimates obtained by the filtering, squaring, and averaging operations

illustrated in Figure 5.5. This approach evolves from the definition of the autospectral density function given in Section 5.2.3 and was implemented in the past by analog instruments, as well as currently on digital computers using computational procedures involving digital filters. However, as detailed in Chapter 11, most current digital data analysis procedures for the estimation of autospectral density functions use finite Fourier transform computations that evolve from the definition of the autospectral density function given in Section 5.2.2. The finite Fourier transforms are accomplished using a fast Fourier transform (FFT) algorithm and, hence, are commonly referred to as *FFT procedures*. The normalized random error for the spectral estimates obtained using FFT procedures is equivalent to that given in Equation (8.147), but the parameters in the error expression are different. The direct derivation of the normalized random error for autospectral density estimates using FFT procedures may add insight into the spectral density estimation error problem.

Consider the autospectral density function of a stationary (ergodic) Gaussian random process $\{x(t)\}$, as defined in Equation (5.67). Specifically, given a sample record $x(t)$ of unlimited length T , the autospectrum is

$$G_{xx}(f) = 2 \lim_{T \rightarrow \infty} \frac{1}{T} E[X(f, T)^2] \quad (8.151)$$

where $X(f, T)$ is the finite Fourier transform of $x(t)$, that is,

$$X(f, T) = \int_0^T x(t) e^{-j2\pi f t} dt \quad (8.152)$$

Now an estimate of $G_{xx}(f)$ can be obtained by simply omitting the limiting and expectation operations in Equation (8.151). This will yield the "raw" estimate

$$\tilde{G}_{xx}(f) = \frac{2}{T} |X(f, T)|^2 \quad (8.153)$$

with the narrowest possible resolution $\Delta f = (1/T)$.

To determine the variance of this estimate, observe that the finite Fourier transform $X(f, T)$ is defined by a series of components at frequencies $f = k/T$, $k = 1, 2, 3, \dots$. Further observe that $X(f, T)$ is a complex number where the real and imaginary parts, $X_R(f, T)$ and $X_I(f, T)$, can be shown to be uncorrelated random variables with zero means and equal variances. Because a Fourier transformation is a linear operation, $X_R(f, T)$ and $X_I(f, T)$ will be Gaussian random variables if $x(t)$ is Gaussian. It follows that the quantity

$$|X(f, T)|^2 = X_R^2(f, T) + X_I^2(f, T) \quad (8.154)$$

is the sum of the squares of two independent Gaussian variables. Hence from

the definition of Equation (4.16) as applied in Equation (4.37), each frequency component of the estimate $\hat{G}_{xx}(f)$ will have a sampling distribution given by

$$\frac{\hat{G}_{xx}(f)}{G_{xx}(f)} = \frac{\chi^2_2}{2} \quad (8.155)$$

where χ^2_2 is the chi-square variable with $n = 2$ degrees-of-freedom.

Note that the result in Equation (8.155) is independent of the record length T , that is, increasing the record length does not alter the distribution function defining the random error of the estimate. It only increases the number of spectral components in the estimate. If the record length is interpreted as a measure of the sample size for the estimate, this implies that Equation (8.153) produces an inconsistent estimate of autospectral density functions, as previously stated in Section 5.2.2. Furthermore, the random error of the estimate is substantial. Referring to Equations (4.19) and (4.20), the mean and variance of the chi-square variable are n and $2n$, respectively. Then the normalized random error is

$$\epsilon_r[\hat{G}_{xx}(f)] = \frac{\sigma[\hat{G}_{xx}(f)]}{G_{xx}(f)} = \frac{\sqrt{2n}}{n} = \sqrt{\frac{2}{n}} \quad (8.156)$$

For the case at hand, $n = 2$, so $\epsilon_r = 1$, which means that the standard deviation of the estimate is as great as the quantity being estimated. This would be an unacceptable random error for most applications.

In practice, the random error of autospectra estimates produced by Equation (8.153) is reduced by computing an ensemble of estimates from n_d different (distinct, disjoint) subrecords, each of length T , and averaging the results to obtain a final "smooth" estimate given by

$$\hat{G}_{xx}(f) = \frac{2}{n_d T} \sum_{i=1}^{n_d} |X_i(f, T)|^2 \quad (8.157)$$

Since each spectral calculation in Equation (8.157) adds two statistical degrees-of-freedom to the estimate, it follows that

$$\epsilon_r[\hat{G}_{xx}(f)] = \sqrt{\frac{2}{2n_d}} = \frac{1}{\sqrt{n_d}} \quad (8.158)$$

The minimum total record length required to compute the autospectrum estimate is clearly

$$T_r = n_d T \quad (8.159)$$

and the resolution bandwidth of the analysis is approximated by

$$B_e \approx \Delta f = 1/T \quad (8.160)$$

Hence, $n_d = (T_r/T) \approx B_e T_r$, so Equation (8.158) is equivalent to

$$\epsilon_r[\hat{G}_{xx}(f)] \approx \frac{1}{\sqrt{B_e T_r}} \quad (8.161)$$

in agreement with Equation (8.147) where T_r replaces T .

Referring back to Equation (8.155), the sampling distribution for an autospectral density estimate may now be written as

$$\frac{\hat{G}_{xx}(f)}{G_{xx}(f)} = \frac{\chi^2_n}{n} \quad n = 2n_d \quad (8.162)$$

From Equation (4.47), it follows that a $(1 - \alpha)$ confidence interval for $G_{xx}(f)$ based on an estimate $\hat{G}_{xx}(f)$ is given by

$$\left[\frac{n\hat{G}_{xx}(f)}{\chi^2_{n; \alpha/2}} \leq G_{xx}(f) \leq \frac{n\hat{G}_{xx}(f)}{\chi^2_{n; 1-\alpha/2}} \right] \quad n = 2n_d \quad (8.163)$$

As before, if the normalized random error is relatively small, say $\epsilon_r \leq 0.10$, then 95% confidence intervals can be approximated from Equations (8.14) and (8.158) by

$$\left[\left(1 - \frac{2}{\sqrt{n_d}} \right) \hat{G}_{xx}(f) \leq G_{xx}(f) \leq \left(1 + \frac{2}{\sqrt{n_d}} \right) \hat{G}_{xx}(f) \right] \quad (8.164)$$

It should be mentioned that the random errors in spectral density calculations might be somewhat greater than indicated in the foregoing equations, depending on the exact details of the calculations, as discussed in Section 11.5.2.

8.5.5 Test for Equivalence of Autospectra

An estimate $\hat{G}(f)$ of an autospectral density function $G(f)$ will have a sampling distribution that is approximately Gaussian if the number of averages n_d is large, say $n_d \geq 30$. It is shown in Section 8.5.4 that the mean value (assuming no bias) and variance of the estimate are given by

$$E[\hat{G}(f)] = G(f) \quad (8.165)$$

$$\text{Var}[\hat{G}(f)] = \frac{1}{n_d} G^2(f) \quad (8.166)$$

Hence a $(1 - \alpha)$ confidence interval for $G(f)$ based on a measurement $\hat{G}(f)$

may be approximated by

$$\left[\hat{G}(f) \left(1 - z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right) \leq G(f) \leq \hat{G}(f) \left(1 + z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right) \right] \quad (8.167)$$

where $z_{\alpha/2}$ is the 100 $\alpha/2$ percentage point of the standardized Gaussian distribution. To arrive at Equation (8.167), it is assumed that $z_{\alpha/2} \sqrt{1/n_d} \ll 1$, so that

$$\left(1 \pm z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right)^{-1} \simeq \left(1 \mp z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right) \quad (8.168)$$

A logarithmic transformation of the estimate $\hat{G}(f)$ to $\log \hat{G}(f)$ has the effect of producing a distribution that is closer to Gaussian than the original distribution. The sample mean value and variance of $\log \hat{G}(f)$ become

$$E[\log \hat{G}(f)] \simeq \log G(f) \quad (8.169)$$

$$\text{Var}[\log \hat{G}(f)] \simeq \frac{1}{n_d} \quad (8.170)$$

Thus the variance here is independent of frequency. Now, a $(1 - \alpha)$ confidence interval for $\log G(f)$ may be approximated by

$$\left[\log \hat{G}(f) - z_{\alpha/2} \sqrt{\frac{1}{n_d}} \leq \log G(f) \leq \log \hat{G}(f) + z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right] \quad (8.171)$$

This result can be derived directly from Equation (8.167) to provide a heuristic explanation for Equations (8.169) and (8.170). This derivation uses the assumption that $z_{\alpha/2} \sqrt{1/n_d} \ll 1$, so that

$$\log \left(1 \pm z_{\alpha/2} \sqrt{\frac{1}{n_d}} \right) \simeq \pm z_{\alpha/2} \sqrt{\frac{1}{n_d}} \quad (8.172)$$

Consider now two different autospectral density function estimates $\hat{G}_1(f)$ and $\hat{G}_2(f)$ obtained under different conditions—for example, from two different sample records or from two different parts of the same sample record. The problem is to decide whether or not these two autospectra are statistically equivalent over some frequency interval (f_a, f_b) of bandwidth $B = f_b - f_a$.

Assume each of the two autospectral density function estimates is based on a resolution bandwidth B_e , where N_f bandwidths are needed to cover the frequency range of interest. That is,

$$N_f = \frac{B}{B_e} \quad (8.173)$$

Further assume the number of averages for each estimate are n_{d1} and n_{d2} , respectively, meaning that the averaging time (record length) for each estimate may be different even though the resolution bandwidth is the same. From Equations (8.169) and (8.170), the sampling distributions of the logarithm of the estimates in the i th bandwidth are approximated by

$$\log \hat{G}_1(f) = y \left[\log G_1(f), \frac{1}{n_{d1}} \right] \quad (8.174)$$

$$\log \hat{G}_2(f) = y \left[\log G_2(f), \frac{1}{n_{d2}} \right]$$

where $y[\mu, \sigma^2]$ is a Gaussian distributed random variable with a mean of μ and a variance of σ^2 . Now, if the two sample records in question have the same autospectral density function $G(f) = G_1(f) = G_2(f)$, it follows from Equations (8.174) that

$$\log \frac{\hat{G}_1(f)}{\hat{G}_2(f)} = y \left[0, \frac{1}{n_{d1}} + \frac{1}{n_{d2}} \right] \quad (8.175)$$

Hence, from Section 4.6, the statistic

$$X^2 = \left[\frac{1}{n_{d1}} + \frac{1}{n_{d2}} \right]^{-1} \sum_{i=1}^n \left[\log \frac{\hat{G}_1(f)}{\hat{G}_2(f)} \right]^2 \quad (8.176)$$

has a chi-square distribution with N_f degrees-of-freedom. That is,

$$X^2 \approx \chi_n^2 \quad n = N_f \quad (8.177)$$

The result in Equation (8.176) provides a basis for testing the hypothesis that $G_1(f) = G_2(f)$. The region of acceptance for the hypothesis test is

$$X^2 \leq \chi_{n;\alpha}^2 \quad n = N_f \quad (8.178)$$

where α is the level of significance for the test, as detailed in Section 4.5.1

8.6 RECORD LENGTH REQUIREMENTS

The error expressions derived in Sections 8.2–8.5 provide ways to assess the statistical accuracy of various parameter estimates after an experiment has

been completed. It would be desirable if these error expressions could also be used to predict the minimum record lengths required in future experiments to obtain data with a predetermined degree of accuracy. The error formulas do relate the total record length T_r to the random error ϵ_r for each parameter estimate. However, these relationships generally include other factors, primarily the autospectrum of the data, which are usually unknown prior to data collection. If one could assume the data have an autospectrum that can be approximated by bandwidth-limited white noise, then the minimum total record length or minimum number of averages needed to achieve a desired accuracy, ignoring bias errors, would be as shown in Table 8.1. Unfortunately, such an assumption is rarely justified in practice so, in most cases, the relationships in Table 8.1 constitute only crude guidelines.

There is one important parameter estimate where reasonable assumptions often can be made that justify the direct use of a result in Table 8.1, namely, an autospectral density estimate. From Equation (8.147), under the assumptions that the autospectrum is relatively smooth over the resolution bandwidth B_e and the data are Gaussian, it is seen that the random portion of the normalized rms error of an estimate is a function only of the total record length T_r and the resolution bandwidth B_e , which are both analysis parameters and not parameters of the data. The only problem then is to select an appropriate value for the resolution bandwidth B_e . This often can be accomplished when

Table 8.1 Record Lengths and Averages for Basic Estimates

Estimate	Required Total Record Length	Required Number of Averages
$\hat{\mu}_x$	$T_r = \frac{1}{2B_e\epsilon_r^2} \left(\frac{\sigma_x}{\mu_x} \right)^2$	$N = \frac{1}{2\epsilon_r^2} \left(\frac{\sigma_x}{\mu_x} \right)^2$
$\hat{\mu}_x^2$	$T_r = \frac{2}{B_e\epsilon_r^2} \left(\frac{\sigma_x}{\mu_x} \right)^2$	$N = \frac{4}{\epsilon_r^2} \left(\frac{\sigma_x}{\mu_x} \right)^2$
$\hat{\sigma}_{xx} \psi_x$	$T_r = \frac{1}{4B_e\epsilon_r^2}$	$N = \frac{1}{2\epsilon_r^2}$
$\hat{\sigma}_{xx}^2 \psi_x^2$	$T_r = \frac{1}{B_e\epsilon_r^2}$	$N = \frac{2}{\epsilon_r^2}$
$\hat{p}(x)$	$T_r = \frac{1}{WBp(x)\epsilon_r^2}$	$N = \frac{2}{Wp(x)\epsilon_r^2}$
$\hat{R}_{xy}(\tau)$	$T_r = \frac{1}{2B_e\epsilon_r^2} [1 + \rho_{xy}^{-2}(\tau)]$	$N = \frac{1}{\epsilon_r^2} [1 + \rho_{xy}^{-2}(\tau)]$
$\hat{G}_{xx}(f)$	$T_r = \frac{1}{B_e\epsilon_r^2}$	$N = \frac{1}{\epsilon_r^2}$

the data represent the response of a lightly damped mechanical or electrical system, as detailed in Section 8.5.1. In this case, the potential bias error for a selected resolution bandwidth B_e can be predicted by either Equation (8.141) or (8.142). Specifically, using Equation (8.141), the resolution bandwidth required to achieve a desired maximum bias error ϵ_b can be predicted by $B_e = 2\zeta f_n (3\epsilon_b)^{1/2}$, where ζ and f_n are the damping ratio and resonance frequency, respectively, of the lightly damped system. At least lower bound values for ζ and f_n commonly can be predicted based upon past experience to achieve a conservative estimate for B_e , which in turn can then be used to predict the minimum required record length from $T_r = (B_e\epsilon_r^2)^{-1}$, as given in Table 8.1.

The foregoing discussions of record length selections for autospectral density function estimates are particularly important for two reasons. First, the spectrum is the single most important parameter of random data for many engineering applications. Second, spectral estimates are generally the most demanding of the various parameter estimates considered in this chapter from the viewpoint of required record length for a given error. This latter fact is easily verified by comparing the error expression for autospectra estimates to the error expressions for other parameter estimates. It is seen that the denominators of the error terms for other parameter estimates generally involve factors that are larger than the value B_e required for well resolved spectral estimates.

PROBLEMS

- 8.1 Which of the following parameter estimates usually involves a bias error as well as a random error?
 - (a) Mean value.
 - (b) Mean square value.
 - (c) Probability density function.
 - (d) Autocorrelation function.
 - (e) Autospectral density function.
- 8.2 The mean value of a bandwidth-limited Gaussian white noise random signal $x(t)$ is estimated from a sample record of length T . Assume the standard deviation of $x(t)$ is $\sigma_x = 1$ and the standard deviation of the estimate is $\sigma[\hat{\mu}_x] = 0.1$. Determine the normalized random error ϵ associated with estimates of the following parameters from the same sample record.
 - (a) Mean square value.
 - (b) rms value.
 - (c) Standard deviation.
- 8.3 Assume $x(t)$ is a sample record of bandwidth-limited Gaussian white noise where $\mu_x = 0.10$, $\sigma_x = 0.20$, $B = 200$ Hz, and $T = 2$ sec. Compute

the normalized rms error for

- (a) mean value estimates.
- (b) mean square value estimates.

8.4

Assume the probability density function of a Gaussian random signal $x(t)$ is estimated using a window width of $W = 0.25\sigma_x$. Determine the bias error of the estimate $\hat{p}(x)$ at $x = \mu_x + 2.5\sigma_x$.

- (a) by exact calculations.
- (b) using the first-order approximation of Equation (8.82).

8.5

Assume the probability distribution (not density) function of a Gaussian random process $\{x(t)\}$ is estimated by ensemble averaging procedures over $N = 100$ independent sample records. Determine the normalized random error of the estimate $\hat{P}(x)$ at $x = \mu$. Is there a bias error in this estimate?

8.6

Consider two bandwidth limited white noise records $x(t)$ and $y(t)$ of bandwidth B and length T where

$$x(t) = s(t) + n_1(t) \quad \text{and} \quad y(t) = s(t) + n_2(t)$$

Assume $s(t)$, $n_1(t)$, and $n_2(t)$ are mutually uncorrelated with zero mean values and mean square values of $\psi_s^2 = S$, $\psi_{n_1}^2 = N_1$, and $\psi_{n_2}^2 = N_2$. Determine

- (a) the normalized rms error of the autocorrelation estimates $\hat{R}_{xx}(\tau)$ and $\hat{R}_{yy}(\tau)$ at $\tau = 0$.
- (b) the normalized rms error of the cross-correlation estimate $\hat{R}_{xy}(\tau)$ at $\tau = 0$.
- (c) the result in (b) if $N_1 \gg S$ and $N_2 \gg S$.
- (d) the result in (b) if $N_1 = 0$ and $N_2 \gg S$.

8.7

Given the two bandwidth-limited white noise signals defined in Problem 8.6, determine an approximate 95% probability interval for the time delay $\hat{\tau}$ corresponding to the maximum value of the estimated cross-correlation function $\hat{R}_{xy}(\tau)$.

8.8

Let $\lambda(f) = |G_{xx}(f)/G_{xx}''(f)|^{1/2}$ and define λ_m as the maximum value of $\lambda(f)$. Determine the requirements on B_e and T as a function of λ_m if the normalized bias error and the normalized random error of an autospectral density function estimate are each to be less than 5%.

8.9

Consider an autospectral density function estimate $\hat{G}_{xx}(f)$ computed from finite Fourier transform operations and the ensemble averaging procedure described in Section 8.5.4. Suppose a sample record $x(t)$ of

total length $T_r = 60$ sec is divided into 12 independent contiguous segments. Determine the 95% confidence interval for the true value $G_{xx}(f)$ when $\hat{G}_{xx}(f) = 0.30$ units²/Hz. What is the normalized random error of the estimate?

- 8.10 Assume a random process $\{x(t)\}$ has an autospectral density function given by

$$G_{xx}(f) = \frac{10}{[1 - (f/100)^2]^2 + [0.1f/100]^2}$$

If the autospectrum is to be estimated from sample data, determine

- (a) the resolution bandwidth B_e and the total record length T_r required to limit the normalized bias error to $\epsilon_b \leq 0.05$ and the normalized random error to $\epsilon_r = 0.10$.
- (b) the resolution bandwidth B_e that will minimize the mean square error of the estimate at $f = 100$ Hz, assuming that a total record length of $T_r = 60$ sec is collected.

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